

How Robotic Artists Perceive ‘Collaboration’ with Machines?

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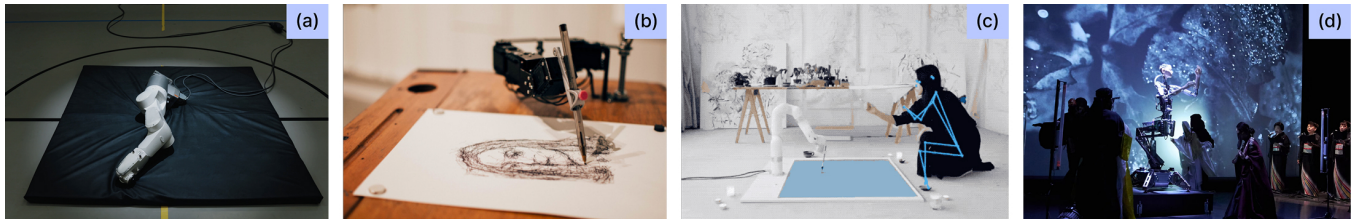


Figure 1: Examples of robotic art: (a) “disarming II” by Emanuel Gollob, a robotic installation to explore the relationships between a detached robotic arm, the environment, and the observers [12]; (b) “Human Study” by Patrick Tresset, a drawing robot with camera depicting live scenes [47]; (c) “Mutations of Presence (D.O.U.G. 4)” by Sougwen Chung, a biofeedback-controlled painting robot [6]; (d) “Alter 3” by Takashi Ikegami, a humanoid robot once in performing Kagura, a Japanese dance ritual [18].

Abstract

This study investigates how artists who build or modify robotic systems perceive human-machine “collaboration” in their artistic practices. Through semi-structured interviews with senior robotic artists, we identify a dissonance between the artists’ perceptions and the narrative of robots as collaborative agents—the artists reject the notion of robots as collaborators, viewing them instead as tools that shape and are shaped by human creativity. Drawing on a materialist perspective, we reflect on the anthropomorphic implication of human-machine collaboration and suggest an alternative framework that recognizes how robots, with their material properties, complement human users in the creative process. The findings contribute to understanding humans’ and machines’ unique contributions and roles in artistic endeavors.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Robotic Art, Robot, Art, Artist, Collaboration, Complementation, Materiality, Agency, Human-robot Interaction

ACM Reference Format:

Yigang Qin, Yanheng Li, and EunJeong Cheon. 2025. How Robotic Artists Perceive ‘Collaboration’ with Machines?. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA ’25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720017>

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CHI EA ’25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720017>

1 Introduction and Related Work

The concept of collaboration arises when discussing interaction between machines and users. For example, human-robot collaboration (e.g., [4]) and human-AI collaboration (e.g., [19, 32]) have gained increasing attention in HCI. This widespread use of the term may be attributable to HCI’s lasting tradition of studying mixed-initiative systems [17, 20], wherein both users and machines contribute to the work, avoiding reliance solely on either direct manipulation by users or full automation by machines [35, 36]. More current designs of human-computer interaction for creativity also relate to this tradition. Examples include systems that take turns with users to complete creative tasks [8, 26, 27]. The focus has been on creating agent-like systems capable of proposing ideas or solutions for specific tasks and situations [5, 38]. Co-creativity systems are especially associated with a mixed-initiative relationship with users [29, 33, 40].

This tradition fuels the idea of human-machine *collaboration* to the extent that we often take collaboration for granted when considering various forms of *interaction*. For instance, a taxonomy of robots’ roles in group work classifies robots as followers, peers, and leaders to human users, all of which terms are considered collaborators of some sort [45]. Some describe machines as cooperative assistants, also mirroring human roles [22]. However, the term *collaboration* has often been applied broadly—or even vaguely—to describe interaction, glossing over whether a genuinely collaborative relationship exists [21, 37]. There is a tendency to equate collaboration with interaction or mutual communication, which has become part of the normative discourse in HCI.

The term *collaboration* can be tricky when describing interactions because it suggests treating machines as cognitively capable as users, which is inappropriate for understanding the existing computing systems. Collaboration, in the strict sense, can be considered as narrowly apt for fully cognitively capable systems, which are not yet feasible. Importantly, the relationships between machines

and users extend beyond what collaboration entails. Machines do not have to be cognitively capable of being useful or meaningful to users. Focusing solely on the term collaboration in research and design can impede us from understanding the full spectrum of relationships between machines and users.

The collaboration framework leads to other implications that we must be concerned about. In a recent study of collaborative robots in the workplace, Cheon et al. [4] demonstrated how the term collaboration could fail to capture the full scope of interactions and tensions between workers and collaborative robots. Genuine collaboration, in many cases, may not exist at all. There are also ethical bearings to this term—it renders machines as something given, essential, and apolitical but overlooks the fact that machines are designed and built through human labor; as Sarkar [43] said, if there is collaboration, it should be the collaboration with the people behind the systems. Moreover, the term conveys an anthropomorphic connotation, which has been particularly prominent in perceptions of the physical, embodied form of machines—robots, which are often seen as equal counterparts to users [7, 23, 30, 34, 41, 44]. Viewing robots as cognitively capable of performing collaborative tasks is related to our tendency to perceive their identities and bodies as a personified whole rather than deconstructing them into “assembled and interconnected technological objects” [13]. As Lucy Suchman said, interactivity with machines increasingly blends more anthropomorphic flavor over its development; however, the resources and ways of learning for humans and machines are essentially different [46].

On the other hand, conservatively speaking, the instrumental view on the ontology of robots is supported by the fact that robots, even empowered by the latest AI, do not possess general intelligence. They are multi-use tools and artifacts to serve human purposes as defined by the EPSRC principles of robotics [2]. The instrumental view is likely a crucial motivation underlying much development in robotics but less so in human-robot interaction for its disciplinary interests. Contrasting these two views in greater detail is informative but beyond the scope. As an early inquiry into this question, the study described in this article shows a group of robotic artists’ take on their relationships with robots in art practices.

Our interest in examining collaboration came from our recent qualitative study with robotic artists on their use of robots in art production [39]. The initial theme of the study focused on robotic art practices and the creativity involved in those practices. Figure 1 shows some examples of robotic artwork, though they do not necessarily come from the artists we engaged with. However, as the study continued, the participants challenged our implicit acceptance of the possibility of a collaborative relationship between the artists and their robots. The challenges happened even when obtaining consent for interviews—one artist initially refused to participate, citing that our participant information sheet inadvertently portrayed robots as collaborative entities in art production, with which they disagreed. Experiences like this made us pause and engage with the concept of collaboration more attentively and raise the following question: *How do robotic artists perceive the idea of collaboration with robots in art production?*

The interview analysis reveals that robotic artists, whose work involves robots participating in the creation or exhibition of artwork, usually see their relationships with the robots as different from a

“collaboration” between humans and machines. Collaboration is a distorted image of the reality of their art practice. The artists highlighted the anthropomorphic connotation implied by collaboration and how it shaped our ideas about what robots can do for creativity. Instead, machines serve as complements—entities enabling human creativity—for them to produce artwork. The inclination towards anthropomorphism in human-machine collaboration may restrict our conceptualization of machines within the confines of human counterparts performing tasks as humans do. Instead, we propose conceptualizing machines in a complementary relationship with human users in creative activities. As Sarkar [43] has argued, we should deconstruct the metaphors of AI being collaborators and partners to revert to the instrumental nature of AI. We should interrogate the metaphor by asking “*Where does it come from? What is it comparing against? If there are so many problems with it, why does the community use it in the first place?*” [43, p.3]” Adopting accurate terms is important as inaccuracy can distort or obscure the phenomena under study and produce designs and deployments that could have been more appropriate.

2 Research Design

2.1 Participant Recruitment

In this study, we conducted semi-structured interviews with nine artists working on robotic art. They are renowned professional artists who use or build physical robots for/their artistic work. Designers and amateurs are out of this criterion. We identified potential participants by browsing individual and artistic organizations’ websites, contacting them directly, and then using snowball sampling to recruit more artists. The recruitment ended when iterative analysis reached saturation. Most participants have had formal art training in drawing, painting, installation, media art, music, sculpture, or dance, and many have a doctoral degree in art or technology. All of them build robots or modify out-of-shell robots. The form of robots ranges from robotic arms and mobile robots to biorobots and humanoid robots. Table 1 shows relevant demographic information of the artists. The study is approved by the Institutional Review Board of the lead author’s institution.

2.2 Semi-Structured Interviews

Before the interviews, the first author reviewed the media press on the artists’ websites, gaining background information to build rapport, develop interview questions, and inform analysis. He conducted the interviews via Zoom with video recording, which lasted an average of 67 minutes. We structured the initial interview questions around the workflow, motivation, rewards, and challenges of working on robotic art, the relationship between artists and robots, and the source of creativity. We aimed to understand their motivations for engaging with robotic art to unpack values embedded in their practices.

The interviews began with the first author asking the participants to *introduce themselves and their work*. Then he asked about *their inspiration source for their work and motivations for working on robotic art*. Following that, he solicited reflections and anecdotes on the *opportunities and challenges of working on robotic art*. He then guided the conversation to explore the *relationships between artists and robots*, focusing on their respective agencies and contributions

Participant	Gender	Age	Geographical Base	Occupation	Years as Artist
Alex	Man	30+	East Asia	University & art	10+
Daniel	Man	30+	Australia	University & art	10+
David	Man	50+	Europe	Full-time art	30+
Evelyn	Woman	50+	North America	Full-time art	10+
Linda	Woman	30+	North America	Non-profit & art	10+
Mark	Man	40+	Europe	Full-time art	10+
Robert	Man	60+	North America	Full-time art	30+
Samuel	Man	60+	East Asia	University & art	20+
Sophie	Woman	30+	Europe	Full-time art	10+

Table 1: Participant demographics in alphabetical order of the pseudonyms.

to creative outcomes. Additionally, he posed questions regarding the *embodied/material engagement* in the art practices. To comply with the semi-structured nature, he prompted follow-up questions throughout the interviews and remained open to new ideas introduced by the participants. After each interview, the participant received a \$40 Amazon or Visa e-gift card if they chose to accept compensation.

2.3 Qualitative Analysis

Data collection and analysis proceeded in parallel and iterative ways throughout the study. We began the analysis shortly after conducting each interview. Emerging insights led to refining the interview protocol to focus on the most salient story in the data. The analysis also informed theoretical sampling [3] at the later stage, which prompted us to purposefully recruit participants who are likely to complete or refine the emerging narrative.

The analysis followed Charmaz’s constructivist grounded theory approach [3] and used Atlas.ti. The first author began with line-by-line open coding on the interview transcripts, followed by focused coding to synthesize and organize the open codes into categories. He compiled analytical memos after each coding phase. The process was iterative and collaborative with other authors, leading to 101 codes in 9 categories. Our preliminary analysis drawing from the RELATIONSHIP category broadly concerns how the artists perceive the relationships of their robots to themselves. For instance, the code “*RELATIONSHIP: instrument*” reflects how the artists perceive robots as tools and instruments for their artistic pursuits; the code “*RELATIONSHIP: impression of agency*” marks conversations about how robots give people the impression of being agentic. We then assigned high-level theoretical codes to the categories to delineate the relationships among the categories. The analysis reached saturation as participants recruited later, to a large extent, repeated previous artists when answering questions regarding our foci.

Our positionalities shaped the analysis. The first and third authors have limited experience in professional art practices, which might hinder their ability to grasp the nuances of the artistic practices fully. The second author, with an educational background in design and arts and publications in art exhibitions, brings an artistic understanding to the analysis.

3 Findings

In our analysis, a predominant sentiment emerged among the artists—they expressed skepticism about portraying their relationship with robots as collaborative. These artists engaged with robots as integral elements of their creative processes, yet they unanimously stopped short of recognizing these machines as *collaborators*. The artists’ hesitation underscores the misalignment of our initial framing with their perceptions of their robots. Rather than viewing these machines as partners, our participants saw them as instrumental complements to their artistic pursuits. However, the machines are not merely facilitative but also components and sites of artistic expression. This complementary relationship suggests an alternative to the narrative that characterizes human-robot interactions as collaborative.

3.1 Questioning Robots as Collaborators in Creative Production

Our participants critically evaluated the concept of human-robot collaboration within the artistic domain. They underscored the indispensable role of human creativity and agency, positing that robots’ current capabilities fall short of the necessary conditions to be qualified as genuine co-creators in the artistic process. According to their responses, robots lack inherent creativity, making the artists attribute artistic merit solely to human inputs.

Alex is an artist who has developed autonomous robots to simulate group behaviors. He viewed these robots primarily as a source of artistic inspiration rather than as collaborative partners. According to Alex, the creative agency in this intersection of art and robotics rests solely on humans; the robots themselves lack creativity and function merely as tools or raw materials in the creative process: “*My robots are just like the physical interface that inspires [me], but not smart enough, not really creative or anything... It’s because they’re not human or the creativity [is] not from them. [They are] made to do [something] by me.*” His perspective underscores a prevalent view of robots as instruments devoid of intentionality, serving merely as conduits for human creativity to be released.

As an artist who used controllers to direct robots equipped with brushes, Evelyn held the view that these robots merely extend her human abilities without possessing agency of their own: “*Everything that’s seen in [my robotic art] is kind of reflecting my abilities, how I can manually operate the robots... They’re an extension of my*

arm... They don't have any agency in [my] works. Every single move comes from me." Evelyn manually manipulated the robots to paint on canvas, ensuring that all artistic decisions originated directly from her as if she was holding a brush. This does not mean that robots are identical to brushes; instead, it underscores that artists maintain a deterministic role in creating art.

Human creativity and agency are dominant not only in utilizing robots but also in designing them. Daniel described how he established the framework and goal of his work, even as he gave the robots greater "freedom"—valuing the unplanned and unexpected behaviors that emerge during artistic practice. Still, he was the one who made everything possible. Even David, who drew together with his robot on canvas, refuted the presence of collaboration with it: "I do a line drawing and then [the robot] completes [it] by doing the shading. But [this situation is] not collaboration. It looks like a collaboration, but [the robot is] just a thing reacting to what I do."

In this case, David and his robot were indeed working together toward a common goal, which is essential for collaboration. However, David saw this condition as insufficient because the robot's shading movement was only a reaction to him rather than a self-determined action. To David, intention is necessary to be creative and collaborative, which robots are yet to possess: "to have creativity, you need the intention of being creative... [Unless] you have an artificial consciousness, you can't have creative [artwork]." Without intention and the creativity that derives from it, the idea of robots as collaborators in art production was difficult for him to accept.

Evelyn and David also mentioned that some artists use the word collaboration only for composing the artist's story and marketing, not necessarily representing their authentic opinions: "I built such a story [of robots being artists] around my relationship with that robot... [It's] a story, I'm not crazy." Evelyn disclosed that she was more frank about this in our interview than in interviews for media presses. Stories can be important for artists to get publicity, but that may not be their genuine opinions.

3.2 Anthropomorphism Behind Collaboration

What accounts for its prevalence if collaboration between humans and robots in creative production is deemed inappropriate? The analysis shows how the term "collaboration" is associated with anthropomorphism. David highlighted that the issue with the term "collaboration" lies in its implication of the autonomy of robots. However, in his artistic practice, the appearance of collaboration was merely a "trick" crafted by the artist:

The problem of using those kinds of words [like collaboration] with robots is that [those words] make [robots] seem more autonomous than they are. Although [robots' autonomous action] is in my installations and [I] make [the robots] look like [how] we behave [as] humans, it is art and it has to be clear that it's [a] theater... [For] the drawing, [collaboration is] a parallel trick... [Robots] can give the impression of being creative, but we have to know that it's not true.

The collaboration we have been discussing may not necessarily reflect the inherent ability of robots but may reflect the impression of autonomy and agency stimulated by robotic systems. Alex, for example, would try to make robots appear autonomous and agentic.

However, that impression and perception of collaboration do not realistically represent the quality of robots. As David pointed out above, artworks are theaters that create impressions for the audience, reflecting how artists utilize this art property to render their work's anthropomorphic image. Similarly, Daniel suggested that humans project meanings onto robots, creating the impression of agency. Sophie attributed this impression to humans' tendency to see something as reflecting themselves: "It [anthropomorphism] is something that as a human we've always done to try to create Frankenstein, to create a figure that represents who we are and [something] we can reflect ourselves into. It's a long tradition in humanity."

Interestingly, Mark explained this anthropomorphism in terms of the specific ways of engaging with robots. He distinguished his way of creating the robots from the audience's way of observation. Building the robots made him perceive them as a running assembly of mechanical components, hardware, and software, which lack agency, while the audience tends to perceive them as completed autonomous moving, living objects:

When I'm working on a project, obviously I spend a lot of time working on it because I have to program and build the object. So I think it's creating [for me] a very different relationship with the object than the people who just discovered the result and the moving object with its living qualities. So for me, at the beginning of the project, the robot itself is just an amount of technical devices, pieces of aluminum or plastics, motors, and the programming environment, so it's not so living. And I'm still influenced [by that initial impression] all the time when I'm working on the project.

The creator can reduce the robot's actions or behaviors to mechanisms of the components, but those who only have access to the final presentation of work cannot. Mark saw moving robots as assemblages of working components rather than independent, autonomous entities. Part of the reason the audience has that impression is that the robots' interactive appearance makes them seem agentic. Therefore, as the creator, Mark is less likely to anthropomorphize robots as agentic, creative, or collaborative counterparts to humans. During our interview, Mark used the essay *On the Marionette Theatre* by Heinrich von Kleist about puppets and puppeteers to make an analogy to robots and robotic artists, showing that creators and audiences have distinct perceptions of robots. Anthropomorphism is rooted in our minds and could be one of the reasons driving the idea of human-machine collaboration.

3.3 Robots as Creative Complements

The next logical question is, what is the alternative to the idea of collaboration that may better capture the mode of interaction between humans and robots in robotic art practices? Our artists directed us to view robots as tools and instruments that complement human creative abilities. Linda used metaphors and analogies—a robot is like a dice to make design decisions and a notebook to record ideas—to describe the role of robots and her relationship with them; for example, she said, "but [using a notebook] doesn't mean I think that my notebook is a collaborator, but I couldn't create what I create without a notebook." Working at the intersection of

robotics and dance performance, she essentially saw robots as tools that complement her abilities to achieve something difficult by herself alone:

It [a robot] is a tool to create, a creativity tool, a creativity assistant, a process you embed yourself in to create things that you wouldn't have otherwise. So if I work with a robot and I roll around on the floor with it, and that comes into being part of my piece, great! Would I have made that without the robot? Absolutely not. It's a tool for creativity. I completely agree. But it doesn't make it a *creator*... the point should be to think about the human and the AI tool as a pair and that the output of the two should be designed with one another in mind.

Linda proposed seeing robots and humans in creative activities as pairs and complements. In contrast, collaboration is a narrower and more special relationship that implies both sides' agency, creativity, and contributions. The same applies to David, who emphasized that the artworks cannot be created solely by either side but rejected the idea that collaboration exists.

Sophie used a brush to make an analogy and showed that robots are helpful tools for creative activities. Though with more complicated functions than brushes, for Sophie, robots are essentially tools that expand her creative ability, extending her body and brain: *"I would say machines are creative as my brush is creative... And with the help of this tool, we create creative elements and we give it agencies... Of course, it [the robot] has a much higher level of freedom and a higher level of decision-making. And this is what I enjoy, I really enjoy exploring this tension, but for itself, it's not creative...I want to use those machines as a way to expand my ability of being creative and to look at that into a new extension of my body and my brain."*

Sophie saw robots as helpful instruments for expanding her creative abilities. Robots, with their unique instrumental affordances, complement humans in creative processes. David similarly regarded robots as *responsive instruments*; only by interacting with them can he create artwork, but they are not necessarily collaborators. However, for David, the current focus on robots enhancing human creativity is misplaced as there are other types of complementation: *"And after [people] came to this thing [of] augmenting creativity. I was like, human creativity is quite good as it is. You don't need to augment it. You need to contain it and to find which ideas are good."* Selecting good ideas generated by users can be one way, among many others, for machines to complement human creativity. Avoiding fixing on developing agents that initiate new ideas may lead us away from the anthropomorphic connotation of collaboration to a more nuanced understanding of the relationship.

As we have shown, the assumption of collaboration between humans and robots does not hold in the eyes of our robotic artists, who perceive robots as tools and instruments that complement their creative abilities rather than as creative or agentive collaborators. This result prompts us to reflect on the prevalent assumption of human-machine collaboration in HCI and explore alternative descriptive language. The following section shows that framing robots as complements rather than collaborators may generate diversified conceptualizations of their roles and design opportunities for human-machine interactions.

4 Discussion

4.1 Human-Machine Complementation

As shown earlier, the conventional concept of collaboration has encountered difficulties in describing the relationship between artists and robots due to the anthropomorphic connotation embedded in the term. The anthropomorphism involves projecting social values and assumptions onto robots, thereby treating robots *only* as part of the social realm. However, our artists reminded us that robots or computer programs lack intention and adequate awareness of other entities and actions in the environment [11], meaning they can hardly be true collaborators. Instead, a *complementary* relationship that draws upon embodied and material interactions with robots emerged across the artists' responses. In such a relationship, robots and artists do not produce art in tandem—generating ideas, communicating, executing, and evaluating together at the same space and time—but rather produce through the artists' iterative recognition of the possibilities and constraints of the robots, which is not mix-initiative or bound to a specific space and time. So a complementary relationship in the general sense is similar to "bounded collaboration"—collaboration with robots that *"is either minimally visible or hardly articulated during the actual use"* [4]. In this way, robots enable and complement the creativity of the artists, but the artists are still at the center of the agency that arranges robots' properties to serve a purpose. Drawing on materialist views [24, 28], we discuss briefly in what follows how the material aspect of robotic systems is essential and unique for the complementary relationship that better captures human-machine interaction in creative activities.

4.2 Material Aspect of Complementation

The material aspect is important to account for the relationship in that it shapes practices conventionally thought as purely social—materials are actors with the agency to influence other actors [25]. Materials, part of nonhumans as proposed by Latour [24], mediate and affect social relations profoundly. Latour gives the joke of soldiers on battlefields to illustrate this idea—if we exclude materials (e.g., clothes, weapons) from the analysis of battles, we would be studying naked soldiers fighting with hands, missing critical components that enable, constrain, and mediate the interactions. Studies of materials can start by studying materiality, which, according to Leonardi [28] is the *"arrangement of an artifact's physical and/or digital materials into particular forms that endure across differences in place and time and are important to users."* Based on this definition, materiality only manifests when utilized and is akin to material affordance.

The materiality of robots enabled robotic art practices in a distinct way. Our artists do not use robots as collaborators but as materials and instruments for creative thinking and making. Their practices suggest we attend to the material aspects of robots/agents and how the robots/agents form a complementary relationship with human users in creative processes. Even the definition of human-robot collaboration by Wang et al. [49] points to a complementary relationship in which robots are material and mechanical operators: *"The main objective of the collaboration is to integrate the best of two worlds: strength, endurance, repeatability, and accuracy of the robots with the intuition, flexibility and versatile problem solving and*

sensory skills of the humans. This relationship of complementarity softens the anthropomorphic inclination embedded in collaboration while recognizing the unique contributions of robots' material aspects to creativity. However, it may be necessary to add that taking an account of materiality is not to wholly undermine the social framework from which collaboration derives (i.e., treating robots as social collaborators) but to bring more attention to the material-embeddedness of the human-machine relationship. In fact, the complementary relationship that we suggest is readily compatible with reasonable social attributions of the nature of robots and their relations to humans.

Prior studies have shown similar results in how material and embodied aspects of robots facilitate creativity. Embodied robotic motions convey emotional expressions and social cues that may enrich and facilitate creation activities like drawings [1, 10, 14, 42, 48]. Embodiment-related constraints (e.g., the physical limitations of moving robotic arms) can stimulate creativity as they push creators to develop new movements for robots, which confirms the principle that encountering material obstacles can lead to generative processes [15]. When embraced, mechanical malfunctions caused by materiality can also become a source of creativity for robotic art [39]. Similar effects are observed in activities such as art and digital fabrication [9, 16]. To conclude, the effectiveness of materiality in propelling and enriching creative experience necessitates shifting from conceptualizing machines with their materiality as *collaborating* with humans to *complementing* humans.

4.3 Design Implications

This complementary relationship with robotic systems may inspire new designs for creative activities thought of as disembodied in previous studies (e.g., [26, 29, 31, 33, 40]). Designers can exploit embodied computing systems' physical appearances, mechanical functions, and material affordances to inspire creative thinking and making. As some of our artists have mentioned, robots serve them as "physical interfaces" that inspire ideas. Furthermore, materials are not only tools but also sites for creative thinking. With much of the innovation of robotic art coming from pushing the boundaries of materiality, such as Alex's customizing physical components to fit particular needs and Robert's juxtaposing plants, animals, and robotic systems to form installation artworks, paying attention to materials in the creative process can enrich the design space. However, the collaboration framework's inherent emphasis on social and cognitive properties may discount such material innovations.

4.4 Limitations and Future Work

Lastly, we acknowledge the study's limitations. There may be differences between professional robotic art in this study and other creative activities—such as commercial design and amateur creative production—that may be associated with differences in perceptions of robotic systems. Artists who do not have similar technical backgrounds or experiences as our participants may also perceive robotic systems differently. It would also be inappropriate to assert that our set of artists represents the opinions of all robotic artists because there are undeniably artists who accept or strive to collaborate with robotic systems in art production and demonstrate unique values by that approach. In light of the above limitations,

future research can investigate the instrumental and anthropomorphic views in contexts including but not limited to creativity, social communication, and industrial production, as well as for computing systems with different sophistication to set up the ground for synthesis and comparison.

5 Conclusion

In this article, we present a way of conceptualizing human-robot relationships as complementary rather than collaborative in robotic art production, with an emphasis on the materiality of robots, to provoke an open conversation about how we can approach human-machine interaction in creative activities. The case study of robotic art is of value on its own in presenting a perspective alternative to the common framework of human-machine collaboration for creativity. In viewing the rapid reconfiguration of creative work by robots and AI, through this article, we invite HCI researchers and practitioners to further discuss the assumptions, implications, heterogeneity, and boundaries of human-machine collaboration in other contexts to reflect on what we gained and what we missed in the elevation of machines from tools to agents.

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